

NexRay AI

Medical Assistant Platform • Clinical Decision Support Report
10 August 2026 | 08:40 PM

Session ID: 56	Patient: Caleb	Modules: Symptom Checker
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Symptom Analysis

Possible Condition	Confidence
Acute Sickle Cell Crisis (Vaso-occlusive Crisis)	92%
Acute Chest Syndrome secondary to Sickle Cell Disease	78%
Severe Malaria with Hemolysis	54%

■ **Urgency Level: Emergency — Immediate action required**

Recommended Tests

- Complete blood count (CBC) with reticulocyte count and blood film
- Chest X-ray to evaluate for acute chest syndrome
- Blood cultures and malaria rapid diagnostic test (RDT) or thick/thin blood smear
- Renal function tests (urea, creatinine) and liver function tests
- Pulse oximetry and arterial blood gas analysis
- Sickling test/hemoglobin electrophoresis if not previously confirmed

Suggested Treatment

- Oxygen therapy to maintain SpO2 >94% given respiratory symptoms
- Aggressive hydration with IV fluids (avoid over-hydration)
- Adequate analgesia - consider morphine or pethidine for severe pain
- Empirical antimalarial therapy (artemether or artemisinin-based combination) pending test results, given fever and West African context
- Antibiotics if bacterial infection suspected (ceftriaxone covers common pathogens)
- Folic acid supplementation

Next Steps

- Admit for inpatient management given severity and multiple organ involvement
- Establish IV access and begin fluid resuscitation immediately
- Obtain chest X-ray urgently to rule out acute chest syndrome
- Perform malaria testing to exclude or confirm concurrent infection
- Monitor vital signs, oxygen saturation, and urine output closely
- Consider blood transfusion if hemoglobin significantly low or acute chest syndrome confirmed

Summary: This 16-year-old male with known sickle cell disease presents with acute multi-system crisis marked by severe pain, respiratory compromise, jaundice, and fever. Acute sickle cell vaso-occlusive crisis with possible acute chest syndrome is the primary concern. However, concurrent malaria infection must be ruled out given the West African setting and overlapping symptoms. This is a medical emergency requiring immediate admission, aggressive supportive care, and urgent investigations. Prognosis depends on rapid recognition and appropriate management to prevent organ damage.

■ **DISCLAIMER:** This report is generated by NexRay AI and is intended solely as a clinical decision-support tool. All findings, conditions, treatments, and recommendations are AI-generated suggestions and do not constitute a formal medical diagnosis or prescription. The clinical judgment of the attending medical professional must be applied before any action is taken.